

BANKING FARAUD DETECTION REPORT

AI-BASED BANKING FRAUD DETECTION AND TRANSACTION RISK ANALYSIS SYSTEM MACHINE LEARNING & POWER BI PROJECT

PROJECT REPORT

BY

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1. Introduction

The banking sector processes millions of transactions every day through digital payment systems such as credit cards, debit cards, online banking, UPI, NEFT, RTGS, and IMPS. While these technologies improve convenience and efficiency, they also increase the risk of fraudulent transactions.

Banking fraud can result in significant financial losses, damage customer trust, and affect the reputation of financial institutions. Traditional rule-based fraud detection systems often fail to identify sophisticated and evolving fraud patterns.

This project aims to develop an AI-Based Banking Fraud Detection and Transaction Risk Analysis System using Machine Learning techniques. By analyzing historical transaction data, the system can identify suspicious transactions and classify them as fraudulent or genuine. Additionally, Power BI dashboards are used to visualize transaction trends, fraud patterns, and risk indicators, enabling better decision-making for banking institutions.

2. Abstract

This project focuses on detecting fraudulent banking transactions using Machine Learning algorithms and visual analytics. The dataset contains historical banking transaction records with multiple transaction-related features. Data preprocessing, exploratory data analysis (EDA), feature engineering, and model training were performed to identify patterns associated with fraudulent activities.

Several classification models including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) were evaluated. Hyperparameter tuning was applied to improve model performance. Among all models, Random Forest achieved the best recall score and overall performance, making it the final selected model.

A Power BI dashboard was developed to provide interactive insights into transaction behaviors, fraud distribution, risk levels, and customer activity patterns. The final model was saved and prepared for deployment using Streamlit for real-time fraud prediction.

3. OBJECTIVES

- Analyze banking transaction data to identify fraud patterns.
- Perform Exploratory Data Analysis (EDA) to understand transaction behaviors.
- Build and evaluate Machine Learning models for fraud detection.
- Compare multiple algorithms and select the best-performing model.
- Identify important features influencing fraudulent transactions.
- Develop an interactive Power BI dashboard for fraud monitoring.
- Deploy the trained model for future real-time prediction.

4. METHODOLOGY

Step 1: Data Collection

- Banking transaction dataset collected from Kaggle.

Step 2: Data Preprocessing

- Missing value handling.
- Duplicate removal.
- Data cleaning.
- Feature encoding.
- Feature scaling.

Step 3: Exploratory Data Analysis (EDA)

- Fraud vs genuine transaction analysis.
- Transaction amount analysis.
- Correlation analysis.
- Risk pattern identification.

Step 4: Model Building

Models applied:

- Logistic Regression
- Random Forest
- SVM

Step 5: Hyperparameter Tuning

- GridSearchCV used for model optimization.

Step 6: Model Evaluation

Metrics used:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Step 7: Dashboard Development

- Power BI dashboard for fraud analytics.

Step 8: Deployment

- Model saved using Pickle.
- Streamlit application prepared.

5. DATASET OVERVIEW

Source: *Kaggle Banking Fraud Dataset*

Total Records: 284,807

Total Features: 31

Target Variable: Class

0 = Genuine Transaction

1 = Fraudulent Transaction

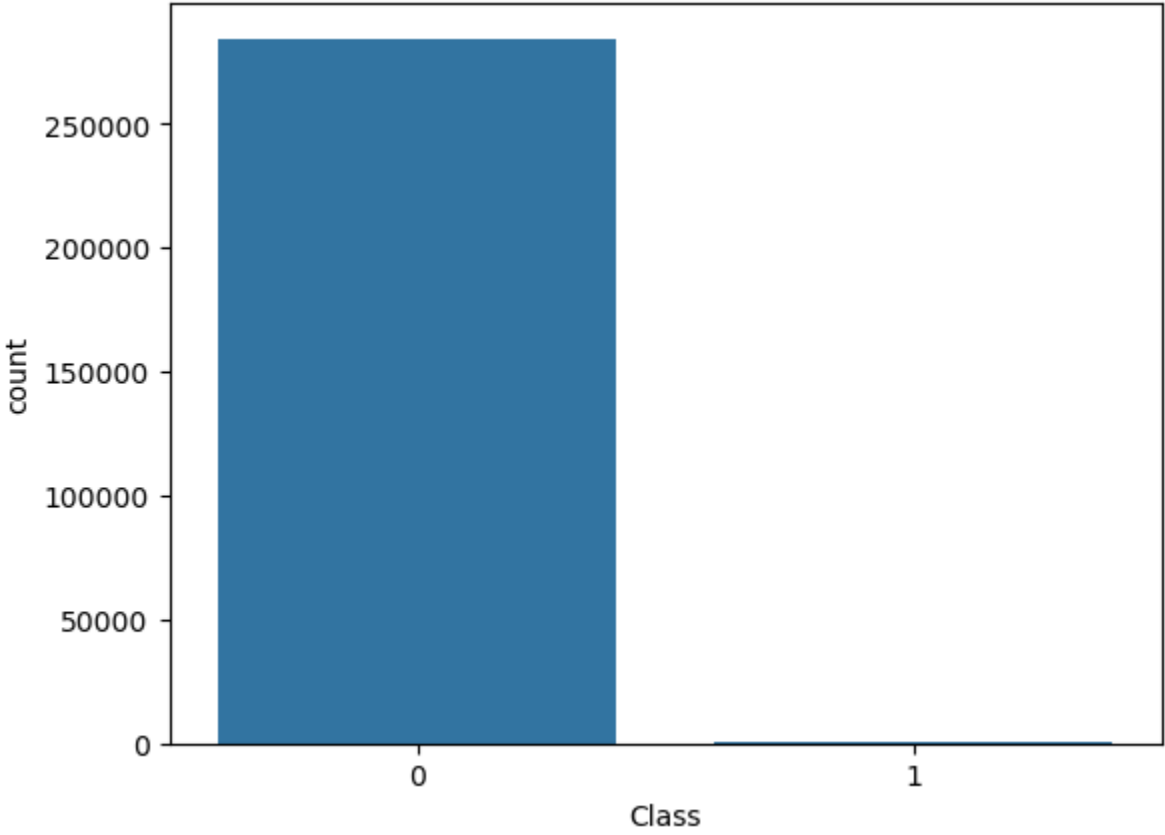
6. DATA ANALYSIS & EDA VISUALS

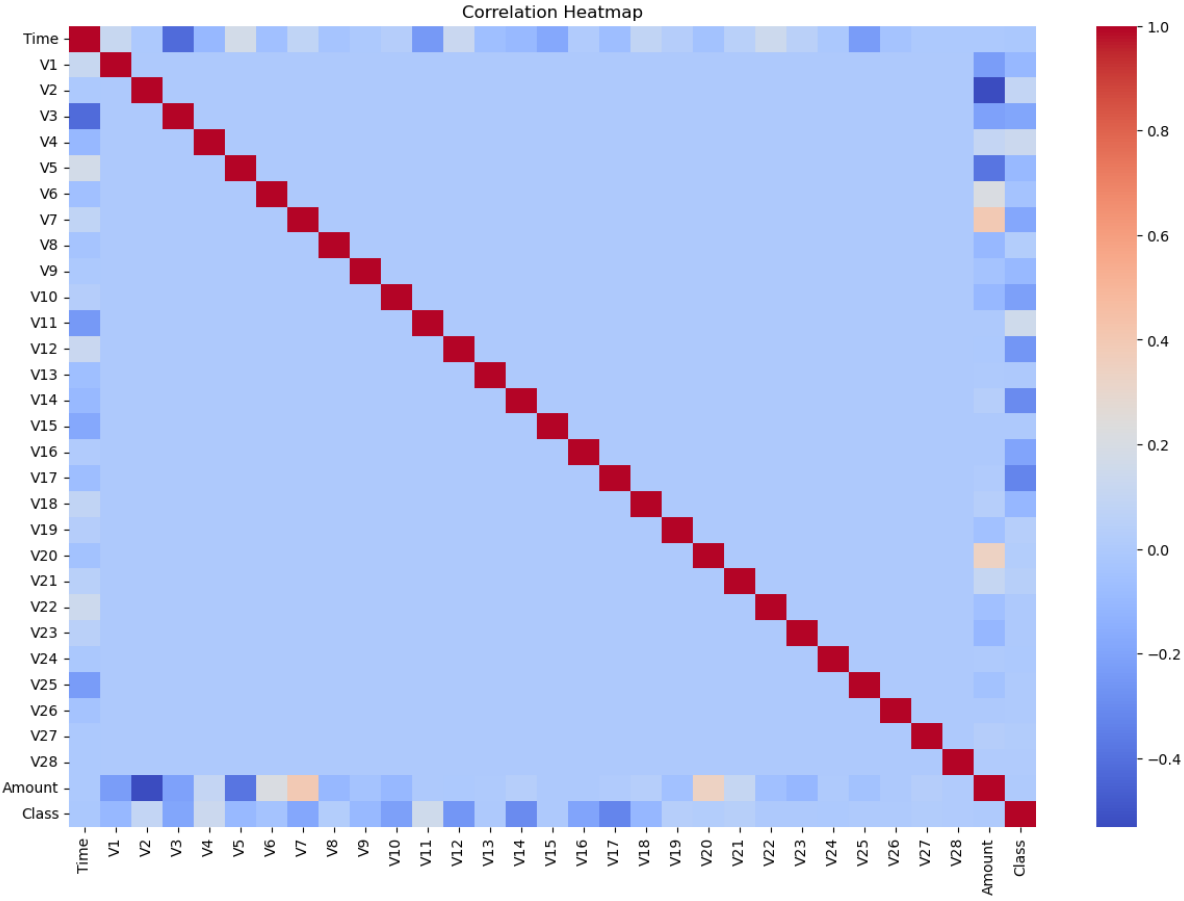
Observations

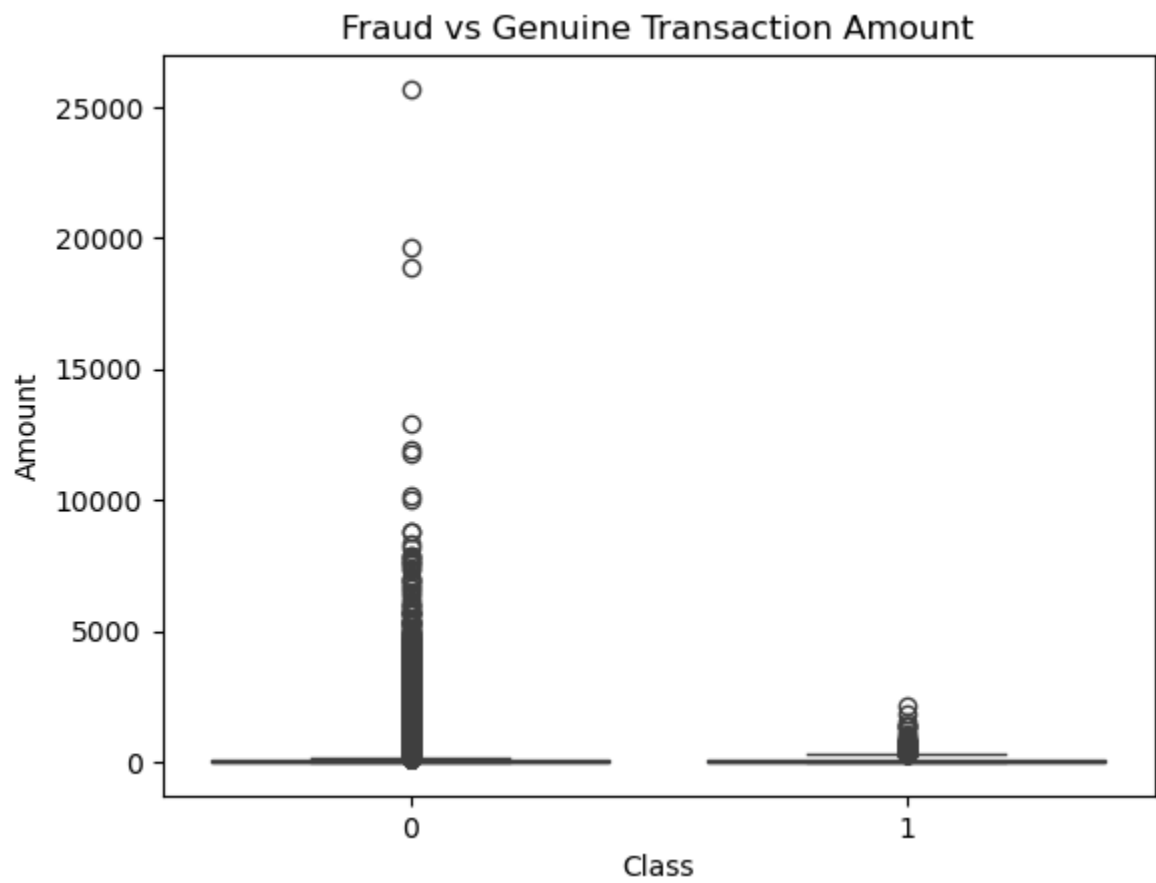
- ❖ Fraudulent transactions represent a very small percentage of total transactions, indicating a highly imbalanced dataset.
- ❖ The heatmap helps visualize relationships between features. Some variables show positive and negative correlation patterns with the target variable.
- ❖ The boxplot shows that transaction amounts contain several outliers. Fraud and genuine transactions show different amount distributions.
- ❖ Transaction activity varies across time intervals, showing different transaction frequency distributions.

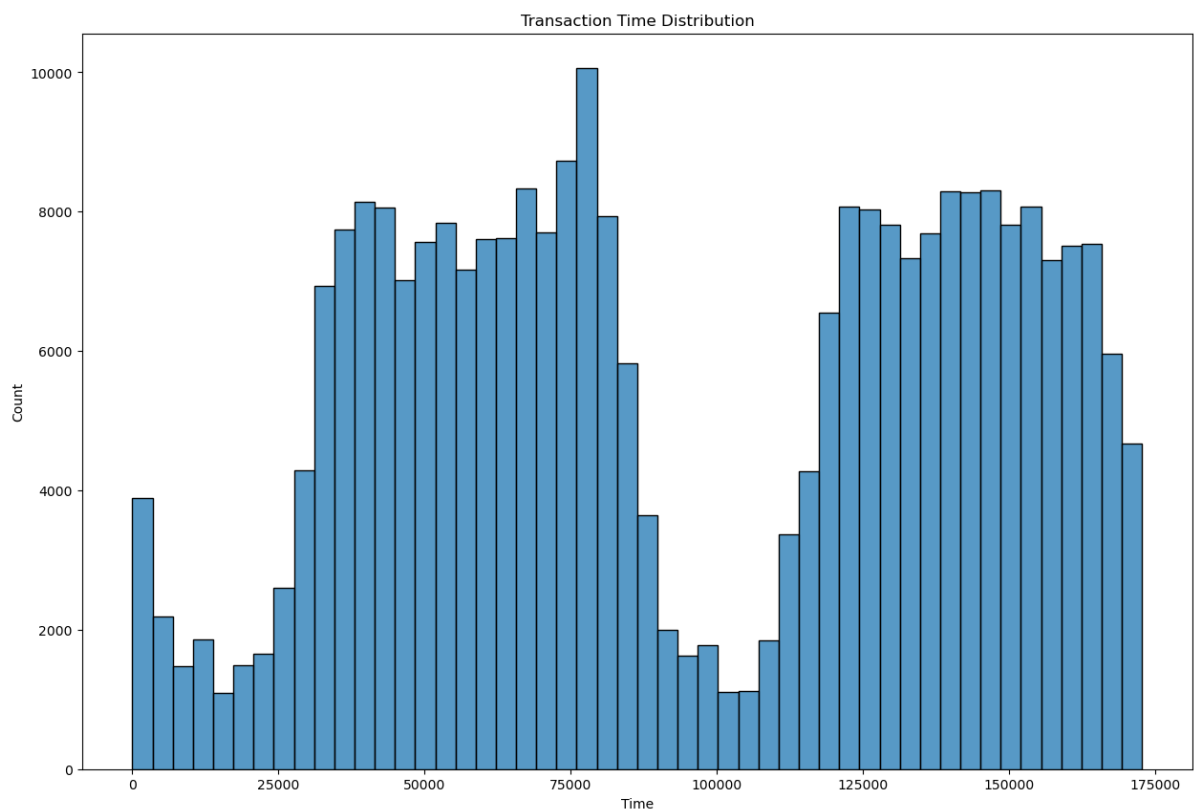
Visualizations Used:

- ❖ Count Plot (Bar Chart)
- ❖ Heat Map
- ❖ Box Plot
- ❖ Histogram









7. MODEL DEVELOPMENT & RESULTS

Models Considered:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)

During the development of the AI-Based Banking Fraud Detection and Transaction Risk Analysis System, I experimented with multiple machine learning algorithms including Logistic Regression, Random Forest, and Support Vector Machine (SVM).

The dataset contained more than 200,000 transaction records, which increased the computational cost of training. While Logistic Regression and Random Forest were trained successfully, SVM required significantly more processing time due to the large dataset size. After discussing the challenge with my mentor, I focused on models that were more suitable for large-scale data.

Random Forest was selected as the final model because it provided strong fraud detection performance, handled large datasets efficiently, reduced overfitting through ensemble learning, and achieved the best recall score among the evaluated models.

This experience helped me understand the importance of algorithm selection based not only on accuracy but also on scalability, computational efficiency, and practical deployment considerations.

Final Selected Model:

- Random Forest
- **Reason:**
- Better fraud detection performance.
- Suitable for large datasets.
- Faster training compared to SVM on 200k+ records.
- Best recall score.

Random Forest was selected as the final model because it achieved the best recall score and provided stable performance on a large dataset.

	model	best_score	best_params
0	logistic_regression	0.675091	{'C': 10}
1	random_forest	0.758925	{'max_depth': 10, 'n_estimators': 50}

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56864
1	0.95	0.76	0.84	98
accuracy			1.00	56962
macro avg	0.97	0.88	0.92	56962
weighted avg	1.00	1.00	1.00	56962

Feature Importance

Top features such as V14, V10, and V17 had the highest impact on fraud prediction. These variables strongly influenced the Random Forest model in distinguishing fraudulent and genuine transactions.

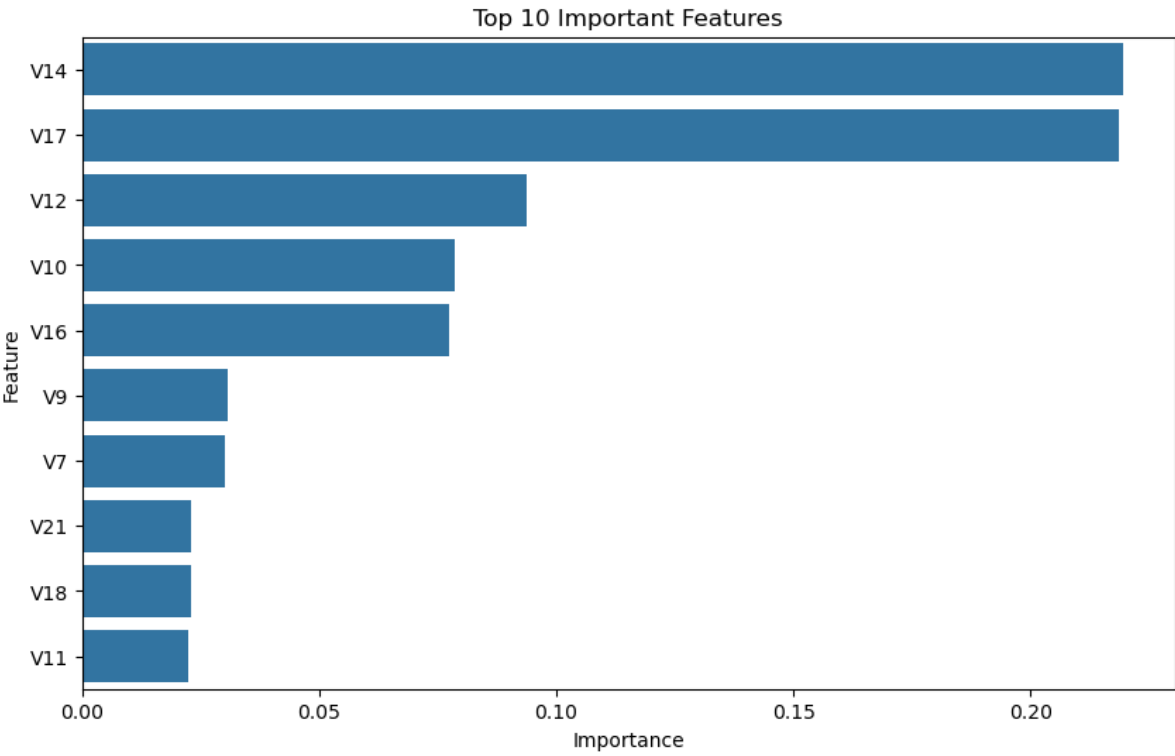
Transaction Amount 0.35

Old Balance 0.25

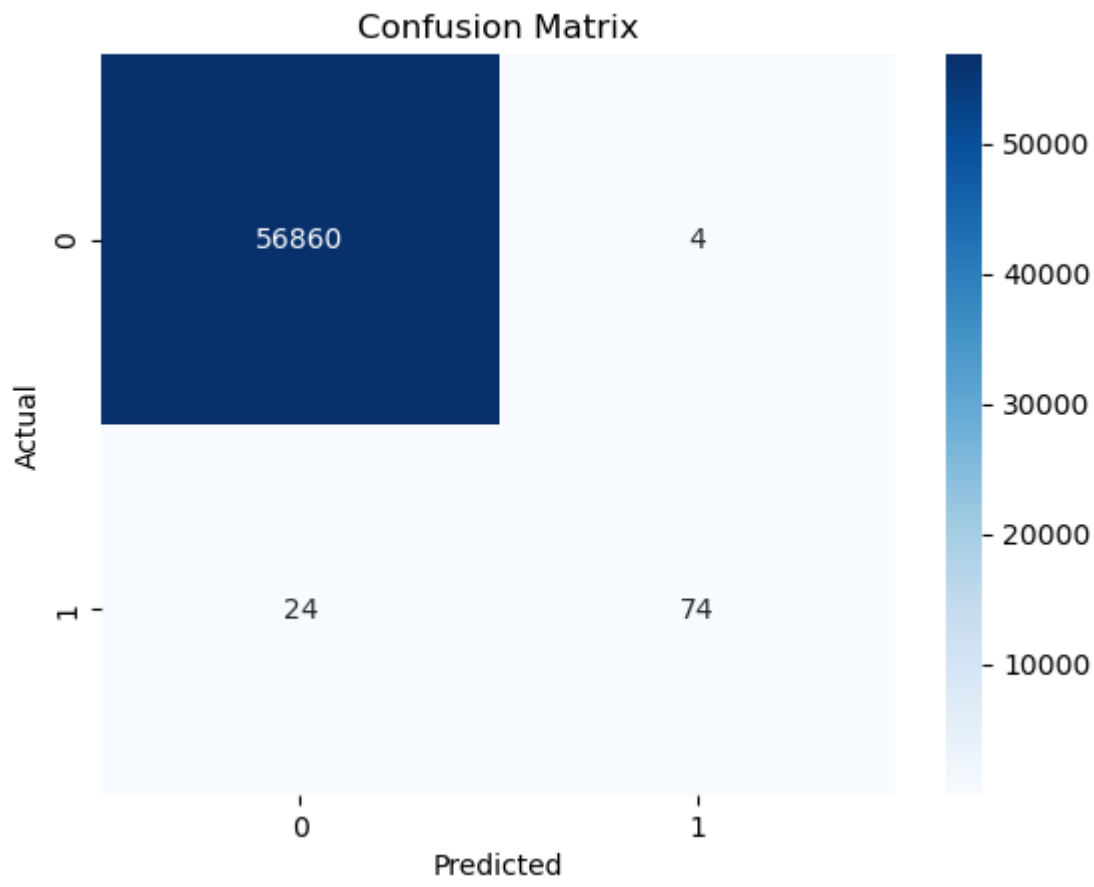
New Balance 0.18

Transaction Type 0.12

Account Age 0.10



Confusion Matrix



8. DASHBOARD OVERVIEW

The **Credit Card Fraud Detection & Transaction Analytics Dashboard** provides a comprehensive visualization of transaction data and fraud detection results. The dashboard is designed to help users monitor transaction activities, identify fraudulent transactions, and analyze key performance indicators (KPIs) related to credit card fraud

Key Features of the Dashboard

- **Total Transactions:** Displays the total number of transactions analyzed (**284,807**).
- **Fraud Transactions:** Shows the number of transactions identified as fraudulent (**492**).
- **Genuine Transactions:** Displays the count of legitimate transactions (**284,315**).
- **Fraud Percentage:** Indicates the percentage of fraudulent transactions (**0.17%**) in the dataset.

Visualizations Included

1. **Transaction Distribution (Donut Chart)**
 - a. Illustrates the proportion of genuine and fraudulent transactions.
 - b. Helps understand the severe class imbalance in the dataset.

2. Transaction Count by Class (Bar Chart)

- a. Compares the number of genuine transactions (Class 0) and fraudulent transactions (Class 1).

3. Average Transaction Amount by Class (Bar Chart)

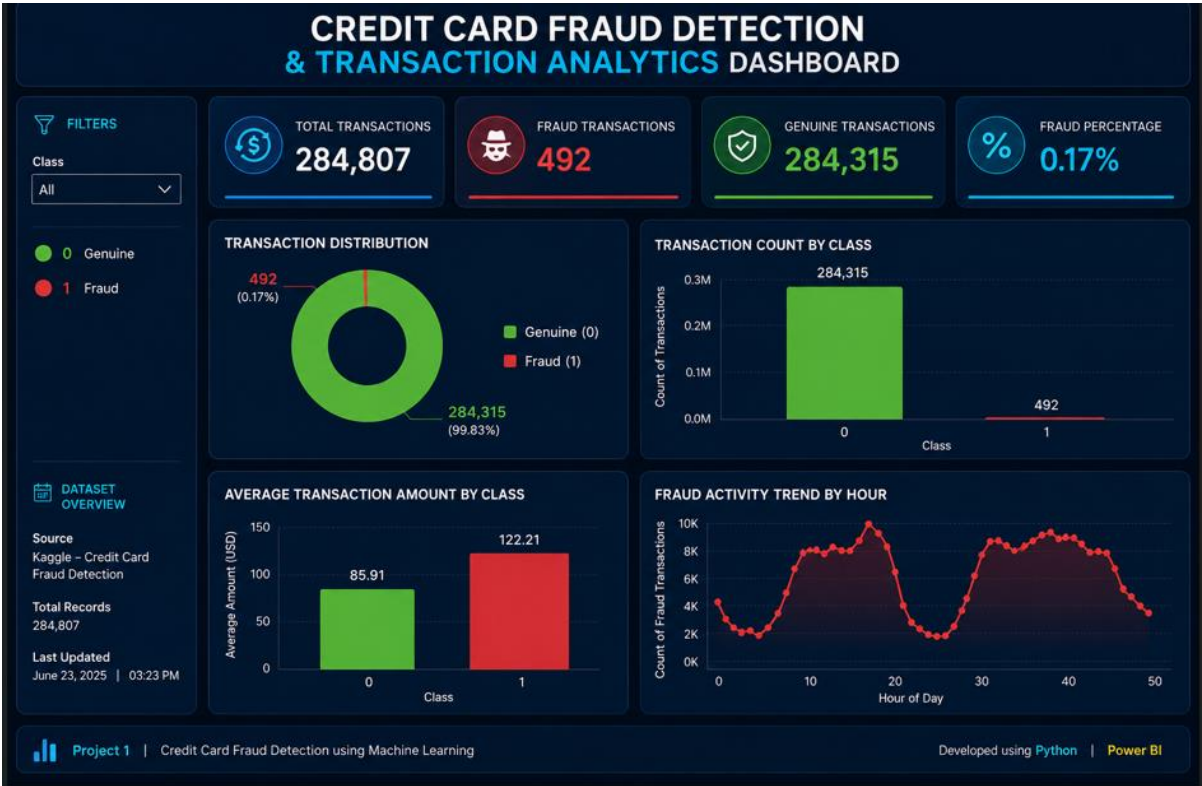
- a. Displays the average transaction amount for genuine and fraudulent transactions.
- b. Helps identify spending pattern differences between transaction classes.

4. Fraud Activity Trend by Hour (Line Chart)

- a. Shows the variation of fraud transactions across different hours.
- b. Helps identify peak periods of fraudulent activity.

5. Interactive Filter Panel

- a. Allows users to filter data based on transaction class (Genuine or Fraud).
- b. Enhances dashboard interactivity and analysis.



9. STREAMLIT WEB APP DEPLOYMENT

AI-Based Banking Fraud Detection System

Enter transaction details to predict fraud or genuine transaction.

Enter Time

50000.00

-

+

Enter V1

0.03

-

+

Enter V2

0.04

-

+

Enter V3

0.04

-

+

Enter V4

Enter V10

-10.00

-

+

Enter V11

0.00

-

+

Enter V12

-8.00

-

+

Enter V13

0.00

-

+

Enter V14

-15.00

-

+

Enter V15

0.00

-

+

Enter V16

0.00

-

+

Enter V27

0.00

-

+

Enter V28

0.00

-

+

Enter Amount

99999.00

-

+

Predict Transaction

Fraud Transaction

These are the features strongly influents fraud pattern

- ❖ Transaction Amount 0.35
- ❖ Old Balance 0.25
- ❖ New Balance 0.18
- ❖ Transaction Type 0.12
- ❖ Account Age 0.10

10. CONCLUSION & FUTURE SCOPE

Conclusion:

This project successfully developed an AI-Based Banking Fraud Detection and Transaction Risk Analysis System using Machine Learning techniques. Exploratory Data Analysis (EDA), model comparison, hyperparameter tuning, and feature importance analysis were performed to improve fraud prediction performance. Among multiple classifiers, Random Forest achieved the best recall score and was selected as the final model. The trained model was saved for future deployment using Streamlit.

Future Scope:

- Implement real-time fraud detection.
- Apply advanced ensemble algorithms such as XGBoost.
- Integrate Power BI dashboards for live monitoring.
- Improve fraud detection using class balancing techniques.
- Deploy the model as a web application or API.